2026

$2 + 0 + 2 + 6 = 10 \rightarrow 1$

$6 - 2 + 0 - 2 = 2$ $\quad$ $026 - 2 = 24$

mod 9 → 1 · mod 11 → 2 · mod 7 → 3 · mod 13 → 11

Numbers That Talk

Divisibility shortcuts, and the modular arithmetic hiding behind them

Full version: proofs, every example, and the theorem map

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MATH 5248 · Fall 2026 · final mini project

What we wanted to know

- ? Why does adding the digits tell you about 3 and 9?
- ? Why does the alternating sum tell you about 11?
- ? Do 7, 13, 37 have rules too? Does every number?
- ? How short can a rule be, and why does 1/7 repeat every 6 digits?

roadmap

1. Foundation: $10 \equiv ? \pmod{d}$
2. Family $10^k \equiv 1$: digit sum, 99, 999 (Zhiheng)
3. Family $10^k \equiv -1$: 11, 101, 1001 (Yueran)
4. Why rules always exist: Fermat, order, 1/p
5. Extras: chop-a-digit, 2 and 5, ISBN

The one idea: what is 10 mod d?

$$n = d_0 + d_1 \cdot 10 + d_2 \cdot 10^2 + \cdots + d_k \cdot 10^k \qquad \text{e.g. } 2026 = 6 + 2 \cdot 10 + 0 \cdot 10^2 + 2 \cdot 10^3$$

Congruence respects + and × (Day 3): if $a \equiv b \pmod m$ then $a^k \equiv b^k$. So we can replace every 10 by whatever 10 is mod d.

10 mod d	after substituting	rule	d
$10 \equiv 1$	$10^i \equiv 1$	add the digits	3, 9
$10 \equiv -1$	$10^i \equiv (-1)^i$	alternate the digits	11
$10 \equiv 0$	$10^i \equiv 0$ $(i \geq 1)$	last digit only	2, 5
$10^k \equiv 1$	treat 10^k as one 'digit'	add k-digit chunks	99, 999, 37
$10^k \equiv -1$	same	alternate k-digit chunks	101, 1001, 7, 13

new

Digit sum: n ≡ S(n) (mod 9)

$$n \equiv d_0 + d_1 + \cdots + d_k = S(n) \pmod 9, \text{ hence also } \pmod 3$$

Proof. $10 = 9 + 1$, so $10 \equiv 1 \pmod 9$. Then $10^i \equiv 1^i = 1$, so $d_i \cdot 10^i \equiv d_i$. Add up: $n \equiv S(n)$. Since $3 \mid 9$, the same holds mod 3. ■

It gives the remainder, not just yes/no, and you can repeat it: 2026 → 10 → 1.



10 steps on a 9-clock = 1 step

7452 → 7+4+5+2 = 18, so 9 | 7452 (= 9 × 828). 2026 → 10 → 1, so 2026 ≡ 1 (mod 9) and (mod 3).

Magic trick 1: the vanishing digit

1. Think of a 4-digit number (not all digits the same).
2. Shuffle its digits into a new number.
3. Subtract: bigger minus smaller.
4. Circle one non-zero digit of the result; hide it.
5. Read out the rest in any order. We name the hidden digit.

5274 → 7425 → 7425 − 5274 = 2151 → hide 5 → hear 2, 1, 1 → sum 4 → 5

8317 → 1378 → 6939 → hide a 9 → hear 6, 3, 9 → sum 18 → 9

why it works

Same digits $\Longrightarrow S(n) = S(n') \Longrightarrow n \equiv n' \pmod 9$.

So the difference $D = n - n' \equiv 0$, and by the theorem $S(D) \equiv D \equiv 0 \pmod 9$.

Hidden digit x, heard sum s: $s + x \equiv 0 \pmod 9$. Exactly one x in 1…9 works.

If s is already a multiple of 9, the answer is 9. Hiding 0 is banned because 0 and 9 would look the same.

If the difference is 0, they shuffled into the same number. Ask again.

new

Bigger chunks: $10^k \equiv 1$

If $10^k \equiv 1 \pmod{d}$ and $n = B_0 + B_1 \cdot 10^k + B_2 \cdot 10^{2k} + \cdots$ (blocks from the right), then $n \equiv B_0 + B_1 + B_2 + \cdots \pmod{d}$

Proof. Writing n in base 10^k and using $(10^k)^j \equiv 1$: the digit-sum proof with 10 replaced by 10^k. ■

12 **34** **53**

k = 2: $100 \equiv 1 \pmod{99}$. 12 + 34 + 53 = 99, so 99 | 123453 (= 99 × 1247); also 3, 9, 11, 33.

45 **658**

k = 3: $1000 \equiv 1 \pmod{999}$, 999 = 27 × 37. 45 + 658 = 703 = 19 × 37, so 37 | 45658 (= 37 × 1234).

116 **667**

116 + 667 = 783 = 27 × 29, so 27 | 116667 (= 27 × 4321). 2026 → 2 + 026 = 28 ⟹ $2026 \equiv 28 \pmod{37}$.

Chunk from the right (45 | 658, not 456 | 58). 3-block sums say nothing about 7 or 13 (they don't divide 999).

new

Alternating sum (mod 11)

Proof. $10 = 11 - 1$, so $10 \equiv -1 \pmod{11}$ and $10^i \equiv (-1)^i$. Substitute into the expansion. ■

$918082 \to 2 - 8 + 0 - 8 + 1 - 9 = -22 = -2 \times 11$, so $11 \mid 918082$ $(= 11 \times 83462)$.



$6 - 2 + 0 - 2 = 2 \implies 2026 \equiv 2 \pmod{11}$

easy to get wrong

For yes/no divisibility you may start from either end.

For the remainder, the + must start at the units digit. 2026 from the left: $2 - 0 + 2 - 6 = -2 \equiv 9$, wrong sign.

A negative result is fine: add a multiple of 11.

$10^k \equiv -1$: alternate the blocks

If $10^k \equiv -1 \pmod d$: $n \equiv B_0 - B_1 + B_2 - \cdots \pmod d$ (same proof, with $(10^k)^j \equiv (-1)^j$)

46 **12** **67**

k = 2: 100 ≡ −1 (mod 101). 67 − 12 + 46 = 101, so 101 | 461267 (= 101 × 4567).

1 **002** **001**

k = 3: 1000 ≡ −1 (mod 1001), 1001 = 7 × 11 × 13. 001 − 002 + 1 = 0, so 7, 11, 13 all divide 1002001 (= 1001^2).

1 **283** **945**

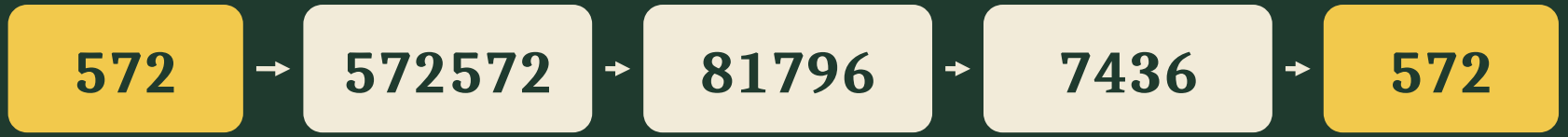
945 − 283 + 1 = 663 = 13 × 51, so 13 | 1283945. And 663 ≡ 5 (mod 7), so 7 ∤ 1283945. Indeed 1283945 ≡ 5 (mod 7).

2 **026**

2026 → 026 − 2 = 24 ⟹ 2026 ≡ 3 (mod 7), ≡ 11 (mod 13), ≡ 2 (mod 11).

Pink block = minus sign. Blocks count from the right; the rightmost block is always +.

Magic trick 2: 1001 brings it back



$$\text{abcabc} = \text{abc} \cdot 1000 + \text{abc} = \text{abc} \cdot 1001 = \text{abc} \cdot 7 \cdot 11 \cdot 13$$

Dividing by 7, then 11, then 13 is dividing by 1001, so you get abc back. The trick cannot fail.

Link to the rule: 572572 → 572 − 572 = 0, so the 3-block alternating sum says 7, 11, 13 all divide it.

Works with a zero too: 205 → 205205 → 29315 → 2665 → 205.

One idea, every rule

find k with $10^k \equiv +1$ or $-1 \pmod{d} \Rightarrow$ add, or alternate, the k-digit blocks

The block sum is congruent to n, so it gives the **remainder**, and you may repeat it until the number is small.

The same idea in base b: digit sum tests b − 1, alternating sum tests b + 1. Octal $144_8 = 100$: digit sum $9 \equiv 2 \pmod 7$ ✓, alternating $4 - 4 + 1 = 1 \equiv 100 \pmod 9$ ✓.

d	k	$10^k \equiv$	rule
3, 9	1	+1	digit sum
11	1	−1	alternate digits
101	2	−1	alternate 2-blocks
27, 37, 111, 333	3	+1	add 3-blocks
7, 13, 77, 91	3	−1	alternate 3-blocks
41, 271	5	+1	add 5-blocks

$99999 = 9 \times 41 \times 271$, $999 = 27 \times 37$, $1001 = 7 \times 11 \times 13$

Every prime p ≠ 2, 5 has a block rule

new

(1) Fermat (Day 11–12): $p \nmid 10 \Longrightarrow 10^{p-1} \equiv 1 \pmod p$. So $(p-1)$-digit blocks, added, always work.

(2) The shortest **additive** block is $k = \mathrm{ord}_p(10)$ (Day 13), and $k \mid p-1$. (A shorter **alternating** rule may exist: 7 has $k = 6$ but uses 3-blocks.)

(3) If k is even, $10^{k/2} \equiv -1 \pmod p$: put $x = 10^{k/2}$. Then $p \mid x^2 - 1 = (x-1)(x+1)$. By Euclid's lemma $p \mid x-1$ or $p \mid x+1$; the first contradicts minimality of k. ■

Careful: (3) needs p prime. For $d = 33$, $\mathrm{ord}(10) = 2$ but $10 \not\equiv -1 \pmod{33}$.

$p = 7$

$10^6 \equiv 1 \pmod 7$ by Fermat

$\mathrm{ord}_7(10) = 6 = p - 1$

6 is even $\Longrightarrow 10^3 = 1000 \equiv -1$

$p = 13$: ord = 6 | 12, $10^3 \equiv -1$

$p = 37$: ord = 3 | 36 (odd: no −1 rule)

$p = 41$: ord = 5 | 40

The table we built

d	ordd(10)	j with $10^j \equiv -1$	1/d (repeating block)	most practical rule
3, 9	1	none	0.(3), 0.(1)	digit sum
11	2	1	0.(09)	alternate digits
37	3	none	0.(027)	add 3-blocks
101	4	2	0.(0099)	alternate 2-blocks
41	5	none	0.(02439)	add 5-blocks
7	6	3	0.(142857)	alternate 3-blocks
13	6	3	0.(076923)	alternate 3-blocks
17	16	8	0.(0588235294117647)	too long → chop a digit
19	18	9	18-digit block	too long → chop a digit

In every row $\mathrm{ord}_p(10)$ divides $p - 1$ (7: 6 | 6; 13: 6 | 12; 37: 3 | 36; 41: 5 | 40), and the block length equals the length of the repeating decimal (next slide).

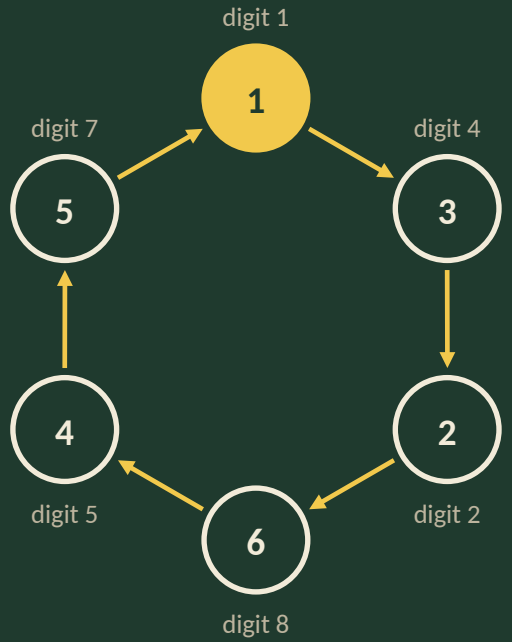
new

Block length = period of 1/d

Theorem. If gcd(d, 10) = 1, d > 1, then 1/d is purely repeating with period $\text{ord}_d(10)$.

Why. Long division: start with remainder $r_0 = 1$; each step multiplies the remainder by 10 and reduces mod d, so $r_i \equiv 10^i \pmod d$. The digits repeat as soon as the remainder returns to 1, i.e. at $i = \text{ord}_d(10)$.

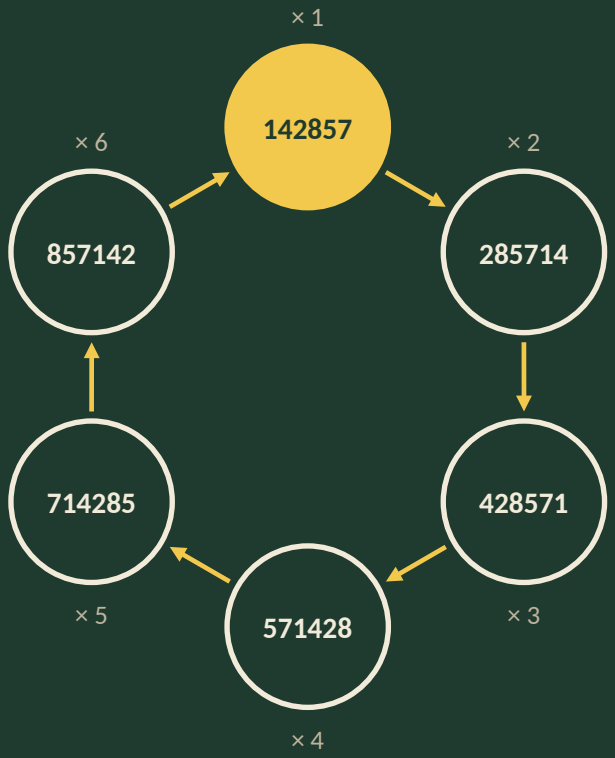
Another view. $10^6 \cdot (1/7) - 1/7 = 142857$, so $999999 = 7 \times 142857$, i.e. $10^6 \equiv 1 \pmod 7$.



remainders of 1 ÷ 7: 1, 3, 2, 6, 4, 5 → back to 1

digit = ⌊10·r / 7⌋ → 1, 4, 2, 8, 5, 7

142857, and Midy's theorem



142857 × 1…6 are rotations of the same six digits; 142857 × 7 = 999999.

Why. 10, 10^2, …, 10^6 mod 7 are 3, 2, 6, 4, 5, 1, every non-zero remainder. So 10 is a **primitive root** mod 7 (Day 14), and 1/7, 2/7, …, 6/7 just start the same remainder cycle at different points.

Midy. For prime p with $10^j \equiv -1$ (period 2j), the two halves of the block add to 99…9: 142 + 857 = 999; 1/13: 076 + 923 = 999; 1/17: 05882352 + 94117647 = 99999999.

Proof sketch: $R = (10^{2j} - 1)/p = (10^j - 1)\cdot c$ with $c = (10^j + 1)/p$ an integer; $R = (c - 1)\cdot 10^j + (10^j - c)$, halves $A = c - 1$, $B = 10^j - c$, $A + B = 10^j - 1$. (p = 7: c = 143, A = 142, B = 857.)

Composite d with gcd(d, 10) = 1

Euler (Day 12–13): $10^{\varphi(d)} \equiv 1 \pmod{d}$, so φ(d)-digit blocks always work; the shortest additive block is still $\text{ord}_d(10)$, which divides φ(d).

d = 27: φ(27) = 18 but $\text{ord}_{27}(10) = 3$ since $999 = 27 \times 37$. $116667 \to 116 + 667 = 783 = 27 \times 29$ ✓

d = 77: $77 \mid 1001$, so $1000 \equiv -1 \pmod{77}$: alternate 3-blocks. $950565 \to 565 - 950 = -385 = -5 \times 77$ ✓ ($950565 = 77 \times 12345$)

CRT view (Day 10–11): mod 77 is mod 7 and mod 11 together, and $10^3 \equiv -1$ on both sides.

any d

$d = 2^a \cdot 5^b \cdot d'$ with $\gcd(d', 10) = 1$

2^a, 5^b: look at the last max(a, b) digits

d′: a block rule from this project

coprime parts ⟹ test separately (next slides)

Chop the last digit: a practical rule for 7

new

$$\gcd(d, 10) = 1,\ m = 10^{-1} \pmod d,\ n = 10x + y \ \Rightarrow\ d \mid n \iff d \mid x + m\cdot y$$

Proof. ($\Leftarrow$) $10(x + my) = 10x + (10m)y \equiv n$. ($\Rightarrow$) $m\cdot n = (10m)x + my \equiv x + my$. Both directions use only that m is the inverse of 10 (Day 4–5). ■

d = 7: $m = 5 \equiv -2$, rule $x - 2y$. 2026 → 202 − 12 = 190 → 19 − 0 = 19, not a multiple of 7 ✓ 343 → 34 − 6 = 28 ✓

d	10^{-1}	rule	example
7	$5 \equiv -2$	x − 2y	2401 → 238 → 7 ✓
13	4	x + 4y	169 → 52 → 13 ✓
17	$12 \equiv -5$	x − 5y	289 → −17 ✓
19	2	x + 2y	361 → 38 ✓

Keeps only yes/no: 2026 ≡ 3 (mod 7) but the chopped 19 ≡ 5. Say 'divisible or not', never 'same remainder'.

2^k, 5^k, and coprime factors

10 ≡ 0. $10^k = 2^k \cdot 5^k \equiv 0 \pmod{2^k}$, so n ≡ (last k digits) (mod 2^k); same for 5^k. 4 and 25: last two digits; 8 and 125: last three.

2026: last digit 6 → even, ≡ 1 (mod 5); last two 26 → ≡ 2 (mod 4); last three 026 → ≡ 2 (mod 8). All match.

Coprime factors. a | n, b | n, gcd(a, b) = 1 ⟹ ab | n. Proof: Bézout gives ax + by = 1, so n = axn + byn, and ab divides both terms. ■

So 6 = 2 × 3, 12 = 4 × 3, 66 = 2 × 3 × 11 are tested piece by piece.

counterexample

$$4 \mid 12,\ 6 \mid 12,\ \text{but}\ 24 \nmid 12$$

gcd(4, 6) = 2, not 1. The right statement is lcm(4, 6) = 12 | n.

Coprimality is what makes the Bézout step work.

In real life: ISBN-10 check digits

$$10\cdot d_1 + 9\cdot d_2 + \cdots + 2\cdot d_9 + 1\cdot d_{10} \equiv 0 \pmod{11} \qquad (d_{10} = \text{X means } 10)$$

0-306-40615-2: 0 + 27 + 0 + 42 + 24 + 0 + 24 + 3 + 10 + 2 = 132 = 12 × 11 ✓

Why 11? It is prime, so every weight 1…10 is a unit mod 11. Change one digit, or swap two neighbours, and the weighted sum changes by a unit times a non-zero amount, never by a multiple of 11 (Euclid's lemma). The error is always caught.

That is a divisibility rule running in reverse: instead of testing a number, you design it so the test passes.

Same ideas, bigger stage: the order of an element and primitive roots are the core of Diffie–Hellman and ElGamal (Day 14–15).

Three things we got wrong at first

"7 has no rule."	It has several: 3-block alternating (1000 ≡ −1), and x − 2y. We had only ever seen the 10 ≡ 1 family.
"The alternating sum can start from either end."	True for yes/no, false for the remainder: 2026 from the left gives −2 ≡ 9, the real remainder is 2.
"The shortest rule is always the order of 10."	The order gives the shortest additive block. If it is even, half of it works with alternating signs, but only for primes (d = 33 breaks it).

One sentence, and one year

If you know what 10 is mod d, you can build the rule for d yourself. Fermat (and Euler) say the rule exists; the order of 10 says how short the additive block can be, and it equals the period of 1/d.

d	rule	on 2026	2026 mod d
2, 5	last digit	6	0, 1 ✓
4, 8	last 2 / 3 digits	26, 026	2, 2 ✓
3, 9	digit sum	2+0+2+6 = 10 → 1	1, 1 ✓
11	alternating sum	6−2+0−2 = 2	2 ✓
7, 13	alternate 3-blocks	026 − 2 = 24	3, 11 ✓
37	add 3-blocks	2 + 026 = 28	28 ✓

2026 = 2 × 1013, and 1013 is prime, so none of these divide it.

Theorem map

from class

T2 congruence respects +, ×, powers (Day 3)

T3 units and inverses mod m (Day 4–5)

T4 Bézout; T5 Euclid's lemma (Day 5–6)

T7 Chinese remainder theorem (Day 10–11)

T8 Fermat's little theorem (Day 11–12)

T9 Euler's theorem; T10 order; T11 primitive roots (Day 12–14)

new in this project

N2 / N3 digit sum and alternating sum, with remainders

N4 / N5 block rules for 99, 999, 101, 1001

N6 / N7 existence via Fermat and Euler; order = shortest additive block

N8 / N9 / N10 period of 1/d, cyclic 142857, Midy

N11 chop-a-digit rule via 10^{-1}; N12 / N13 powers of 2, 5 and coprime splitting

N14 other bases; N15 / N16 the two tricks; N18 ISBN-10

References

MATH 5248 course notes: Day 3 (congruences), Day 4–5 (units, inverses, Bézout), Day 6 (Euclid's lemma), Day 10–11 (CRT), Day 11–12 (Fermat), Day 12–13 (Euler), Day 13 (order), Day 14 (primitive roots).

J. Hoffstein, J. Pipher, J. H. Silverman, An Introduction to Mathematical Cryptography, 2nd ed., Springer. Fermat's little theorem, pp. 30–32.

NTIC, Number Theory: In Context and Interactive (link on the course Canvas page).

Wikipedia: Divisibility rule; Repeating decimal; Multiplicative order; Cyclic number; Midy's theorem; International Standard Book Number.